GCB-Net: Graph Convolutional Broad Network and Its Application in Emotion Recognition

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Abstract—In recent years, emotion recognition has become a research focus in the area of artificial intelligence. Due to its irregular structure, EEG data can be analyzed by applying graphical based algorithms or models much more efficiently. In this work, a Graph Convolutional Broad Network (GCB-net) was designed for exploring the deeper-level information of graph-structured data. It used the graph convolutional layer to extract features of graph-structured input and stacks multiple regular convolutional layers to extract relatively abstract features. The final concatenation utilized the broad concept, which preserves the outputs of all hierarchical layers, allowing the model to search features in broad spaces. To improve the performance of the proposed GCB-net, the broad learning system (BLS) was applied to enhance its features. For comparison, two individual experiments were conducted to examine the efficiency of the proposed GCB-net based on the SJTU emotion EEG dataset (SEED) and DREAMER dataset respectively. In SEED, compared with other state-of-art methods, the GCB-net could better promote the accuracy (reaching 94.24 percent) on the DE feature of the all-frequency band. In DREAMER dataset, GCB-net performed better than other models with the same setting. Furthermore, the GCB-net reached high accuracies of 86.99, 89.32 and 89.20 percent on dimensions of Valence, Arousal and Dominance respectively. The experimental results showed the robust classifying ability of the GCB-net and BLS in EEG emotion recognition.

Index Terms—Emotion recognition, graph convolutional neural networks, broad learning system, graph convolutional broad network

1 Introduction

Human emotion is a complex mental state that is closely linked to the brain's responses to internal or external events. As the foundation of feeling, understanding and affective interaction, emotion recognition is obviously a significant area in the field of artificial intelligence. Generally, emotions can be perceived by means of visual signals (e.g., facial expressions [1], body movements [2] or gestures [3]), audio signals (e.g., speech [4]) or physiological signals (e.g., heart rate (HR) [5], [6], respiration rate (RR) [7], galvanic skin response (GSR) [8], electromyography (EMG) [9] and electroencephalography (EEG) [10]). Given that human can conceal emotions or deceive by controlling internal/external behaviors, physiological signals are usually perceived as more reliable than non-physiological ones.

EEG is widely used for emotion recognition due to its easy accessibility and high accuracy [11]. The ability to record the electrical activities of the brain through multiple electrodes placed on the scalp makes it a noninvasive, convenient and low-cost approach to acquire data. Moreover, with its high temporal resolution, EEG can provide more details for emotional analysis. However, there are some problems with EEG emotion recognition, especially the poor spatial resolution and low signal-to-noise ratio of EEG signals. In addition, the performance of EEG data is usually unstable for emotion recognition, due to the complexity of emotions and individual difference. Moreover, it is hard to describe the relations between EEG signals and active areas of the brain, leading to the difficulty in modeling. Therefore, emotion recognition based on EEG data is still a challenging task.

Research on EEG emotion recognition mainly involves two aspects: feature extraction and classification. And the extracted features of EEG signals can be basically categorized into three kinds: time domain, frequency domain and time-frequency domain. For time domain features, statistical index [12], fractal dimensions (e.g., Higuchi Fractal Dimension (HFD) [13]) and non-stationary index (NSI) [14] and higher order crossings (HOC) [15] are often used. For frequency domain features, the frequently used methods include power spectral density (PSD) [16], approximate entropy (AE) [17], differential entropy (DE) [18] and sample entropy (SE) [19]. And for time-frequency domain features, the common extraction methods are fast fourier transform (FFT) [20], discrete fourier transform (DFT) [21] and wavelet transform (WT) [22].

There have been numerous studies on EEG emotion classification. Literature shows that most of them approached extracted EEG features through conventional classifiers such as support vector machines (SVM) [23], k-nearest neighbors (kNN) [24], linear discriminant analysis (LDA) [25], naive bayes (NB) [26], and other evolutionary algorithms [27], [28]. In recent years, deep learning methods have become prevalent in many areas, and EEG emotion

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Manuscript received 26 Mar. 2019; revised 11 July 2019; accepted 8 Aug. 2019. Date of publication 27 Aug. 2019; date of current version 1 Mar. 2022. (Corresponding author: C. L. P. Chen.)
Recommended for acceptance by B. Hu.
Digital Object Identifier no. 10.1109/TAFFC.2019.2937768

GCB-Net：图卷积广网络及其在情感识别中的应用

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摘要—近年来，情感识别已成为人工智能领域的研究重点。由于脑电图数据的结构不规则，通过应用基于图形的算法或模型可以更有效地分析脑电图数据。在这项工作中，设计了一种图卷积广网络（GCB-net）用于探索图结构数据的深层信息。它使用图卷积层提取图结构输入的特征，并堆叠多个常规卷积层来提取

相对抽象的特征。最终的级联使用了广义概念，保留了所有层级输出的信息，使模型能够在广义空间中搜索特征。为了提高所提出的 GCB-net 的性能，应用了广义学习系统（BLS）来增强其特征。为了比较，分别基于 SJTU 情绪脑电数据集（SEED）和 DREAMER 数据集进行了两个独立的实验，以检验所提出的 GCB-net 的效率。在 SEED 中，与其他最先进的方法相比，GCB-net 在所有频段 DE 特征的准确性上表现更好（达到 94.24%）。在 DREAMER 数据集中，GCB-net 的表现优于其他设置相同的模型。此外，GCB-net 在效价、唤醒度和支配性维度上分别达到了 86.99%、89.32%和 89.20%的高准确率。

实验结果表明，GCB-net 和 BLS 在脑电情绪识别中具有强大的分类能力。

索引词—情绪识别，图卷积神经网络，广义学习系统，图卷积广义网络

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1 引言

人类情感是一种复杂的心理状态，与大脑对内部或外部事件的反应密切相关。作为感受、理解和情感互动的基础，情感识别无疑是人工智能领域的一个重要领域。通常，情感可以通过视觉信号（例如，面部表情[1]、身体动作[2]或手势[3]）、音频信号（例如，语音[4]）或生理信号（例如，心率(HR)[5][6]、呼吸率(RR)[7]、皮肤电反应(GSR)[8]、肌电图(EMG)[9]和脑电图(EEG)[10]）来感知。鉴于人类可以通过控制内部/外部行为来隐藏情感或欺骗，生理信号通常被认为比非生理信号更可靠。由于脑电图易于获取且准确性高[11]，它被广泛用于情感识别。通过头皮上放置的多电极记录大脑的电气活动，使其成为一种无创、便捷且低成本的获取数据的方法。此外，凭借其高时间分辨率，脑电图可以为情感分析提供更多细节。 然而

脑电图情绪识别存在一些问题，尤其是脑电图信号的空间分辨率差和信噪比低。此外，由于情绪的复杂性和个体差异，脑电图数据的识别性能通常不稳定。而且，很难描述脑电图信号与大脑活动区域之间的关系，导致建模困难。因此，基于脑电图数据的情绪识别仍然是一项具有挑战性的任务。

脑电图情绪识别的研究主要涉及两个方面：特征提取和分类。而脑电图信号的提取特征基本上可以分为三类：时域、频域和时频域。对于时域特征，常用的统计指标[12]、分形维数（例如，Higuchi 分形维数（HFD）[13]）和非平稳指数（NSI）[14]以及高阶穿越（HOC）[15]等。对于频域特征，常用的方法包括功率谱密度（PSD）[16]、近似熵（AE）[17]、差分熵（DE）[18]和样本熵（SE）[19]。而对于时频域特征，常见的提取方法有快速傅里叶变换（FFT）[20]、离散傅里叶变换（DFT）[21]和小波变换（WT）[22]。

关于脑电图情绪分类的研究有很多。文献表明，大多数研究通过传统分类器提取脑电图特征，例如支持向量机（SVM）[23]、k 近邻（kNN）[24]、线性判别分析（LDA）[25]、朴素贝叶斯（NB）[26]以及其他进化算法[27]、[28]。近年来，深度学习方法在许多领域变得流行，脑电图情绪

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稿件收到日期：2019 年 3 月 26 日；修改日期：2019 年 7 月 11 日；接受日期：2019 年。出版日期：2019 年 8 月 27 日；当前版本日期：2022 年 3 月 1 日。（通讯作者：C. L. P. Chen。）建议由 B. Hu 接受。

数字对象标识符：10.1109/TAFFC.2019.2937768

recognition is no exception. Zheng et al. [29] applied a deep belief network (DBN) to analyze EEG data. Yang et al. [30] proposed a hierarchical network using subnetwork nodes to recognize EEG emotions. Li et al. [31] first utilized the convolutional neural networks (CNN) to extract features constructed through the spectral energy distribution of multi-channel EEG signals within a certain period of time, and then predicted sequence features with long short-term memory (LSTM).

CNN [32] can extract local stationary features using convolutional kernels, which is a superior ability for processing 1D (e.g., text and audio), 2D (e.g., images) and 3D (e.g., videos) regular grid data. However, topologies in the real world, such as social networks, protein-interaction networks and traffic networks, are not always grid-like. The structure of EEG channel is also irregular, since it cannot be easily regarded as the physical position of electrodes. Though some researches successfully transformed continuous EEG frames to a regular grid for convolution, they ignored the structure information of channels. To tackle this problem, graph convolutional neural networks (GCNN) [33], [34], [35] were introduced to generalize convolution on graph data from spectral domain. In 2018, Song et al. [36] applied GCNN for EEG emotion recognition and obtained sound performance. They then proposed dynamical graph convolutional neural networks (DGCNN) to explore the intrinsic connections between EEG channels and implement the graph convolution operation on learned graphs.

While GCNN cannot stay very deep, its accuracy will decrease when the depth of graph convolution layers increases. With more added layers, the effective context of each node increases, thus making it difficult to train GCNN [37]. Li et al. [38] noted that graph convolution can be viewed as a special form of Laplacian smoothing. In this light, adding layers may cause over-smoothing of learned features and mixing of vertices within different clusters, especially for small datasets.

Based on previous works, it can be inferred that a few layers of graph convolution are adequate to learn the topological features of graph structure, while superfluous graph convolution layers may not obtain more distinguishing representations. Hence the Graph Convolutional Broad Network (GCB-net) was proposed which uses regular convolution to explore deeper-level information. The GCB-net stacks regular CNN after graph convolution to abstract high-level features of learned graph representations. Moreover, the network adopts a broad concept and concatenates different hierarchical features (graph convolution layer and regular convolution layer) before final classification. Therefore, the GCB-net is a model that can extract more discriminative patterns of graph data by adding regular CNN and preserve more information for searching features in broad spaces through layer concatenation.

There are a few reasons for proposing GCB-net. First, only using graph convolution to construct network is simplex to deal with graph-structured data, and the complex model usually has more expressive power to fit possible distribution. Second, broad connection can enhance the stability of model to ensure that performance of the whole network won't be worse than a single hierarchy. Third, graph convolution and regular convolution can be unified for different dimensional data. 1D, 2D or 3D CNN can be used while the dimension of input nodes is 1, 2 or 3.

Recently, the Broad Learning System (BLS) [39], [40], [41] has showed its promising performance on small datasets. Its structure is based on random vector functional-link neural networks (RVFLNN) [42], [43]. The designing process of BLS is as follows. First, inputs are randomly mapped to feature nodes. Then the mapped features are randomly generated into enhancement nodes. And finally, the feature nodes and enhancement nodes are connected and fed into the output. BLS is a kind of flat network with broad random possibility obtained through the expansion of feature nodes and enhancement nodes, so that the model can find suitable features to predict. With its good performance in time and classification, BLS was used in this paper to improve the GCB-net.

Contributions of this paper can be summarized in following aspects.

1) Regular convolution was first used to explore graph convolutional features. GCB-net takes advantage of both graph convolution and regular convolution. Graph convolution helps to learn features on graph-structured data, and stacking of regular convolutional layers makes the network deeper so that can abstract high-level information.
2) The broad approach was applied in the model. GCB-net connects the outputs of graph convolution and regular convolution before final classification. The concatenation of all hierarchical layers provides multi-level features and preserves diversity, so that the model can search desirable results in broad feature spaces.
3) The GCB-net and BLS were jointly used in the experiments. BLS was utilized to enhance feature from GCB-net. In the task of recognizing EEG emotion, GCB-net and BLS achieved preferable results compared with other models, demonstrating the robust classifying ability for GCB-net and BLS in EEG emotion recognition.

2 Preliminary

This section presents the basic theories of regular convolution, graph convolution and broad learning system, which are the rudimentary knowledge for constructing the GCB-net.

2.1 Convolutional Operation

Convolution is an operation involved with two functions (data function and kernel function) and typically denoted as operator $*$. Most neural networks implement convolution without kernel flipping and 1D convolution operation can be simply defined as

$$S(i) = (X * W)(i) = \sum_n X(i+n)W(n), \tag{1}$$

where $X(\cdot)$ is 1D data and $W(\cdot)$ is a 1D kernel. By extension, higher dimensional convolution can be written in similar way, e.g., 2D convolution operation

识别也不例外。郑等人[29]应用深度信念网络（DBN）分析脑电图数据。杨等人[30]提出使用子网络节点构建的层次网络来识别脑电图情绪。李等人[31]首先利用卷积神经网络（CNN）提取多通道脑电图信号在特定时间段内通过频谱能量分布构建的特征，然后使用长短期记忆（LSTM）预测序列特征。

CNN [32] 可以使用卷积核提取局部平稳特征，这对于处理 1D（例如文本和音频）、2D（例如图像）和 3D（例如视频）的规则网格数据具有优越的能力。然而，现实世界中的拓扑结构，如社交网络、蛋白质相互作用网络和交通网络，并不总是呈网格状。脑电图（EEG）通道的结构也是不规则的，因为它不能轻易被视为电极的物理位置。尽管一些研究成功地将连续的 EEG 帧转换为规则网格以进行卷积，但他们忽略了通道的结构信息。为了解决这个问题，图卷积神经网络（GCNN）[33]、[34]、[35]被引入，将频谱域中的卷积推广到图数据。2018 年，Song 等人[36]将 GCNN 应用于 EEG 情绪识别，并取得了良好的性能。随后，他们提出了动态图卷积神经网络（DGCNN），以探索 EEG 通道之间的内在连接，并在学习到的图上实现图卷积操作。

虽然 GCNN 不能非常深，但当图卷积层的深度增加时，其准确率会下降。随着更多层的添加，每个节点的有效上下文会增加，从而使得训练 GCNN 变得困难[37]。李等人[38]指出，图卷积可以被视为一种特殊的拉普拉斯平滑形式。从这个角度来看，添加层可能会导致学习到的特征过度平滑以及不同簇内顶点的混合，特别是对于小数据集。

基于先前的研究，可以推断出几层图卷积足以学习图结构的拓扑特征，而多余的图卷积层可能无法获得更具区分度的表示。因此提出了图卷积广义网络（GCB-net），该网络使用常规卷积来探索更深层次的信息。GCB-net 在图卷积之后堆叠常规 CNN，以提取学习到的图表示的高级特征。此外，该网络采用广义概念，在最终分类前将不同层次的特征（图卷积层和常规卷积层）进行拼接。因此，GCB-net 是一个通过添加常规 CNN 来提取图数据更多判别模式的模型，并通过层拼接保留更多信息，以便在广义空间中搜索特征。

提出 GCB-net 有几个原因。首先，仅使用图卷积来构建网络，处理图结构数据较为简单，而复杂的模型通常具有更强的表达能力，能够拟合可能的分布。其次，广泛连接可以增强模型的稳定性，确保整个网络的性能不会比单一层次更差。第三图卷积和常规卷积可以统一用于不同维度的数据。当输入节点的维度为 1、2 或 3 时，可以使用 1D、2D 或 3D CNN。

最近，广义学习系统（BLS）[39]、[40]、[41]在小数据集上展现了其良好的性能。其结构基于随机向量功能链接神经网络（RVFLNN）[42]、[43]。BLS 的设计过程如下。首先，输入被随机映射到特征节点。然后，映射的特征被随机生成到增强节点。最后，特征节点和增强节点被连接并输入到输出端。BLS 是一种平面网络，通过特征节点和增强节点的扩展获得了广泛的随机可能性，使得模型能够找到合适的特征进行预测。凭借其在时间和分类方面的良好性能，本文使用 BLS 来改进 GCB-net。

本文的贡献可以总结为以下几个方面。

1) 首先使用了常规卷积来探索图卷积特征。GCB-net 同时利用了图卷积和常规卷积。图卷积有助于在图结构数据上学习特征，而常规卷积层的堆叠使网络更深，从而可以提取高级信息。

2) 广义方法被应用于该模型。GCBnet 在最终分类前连接图卷积和常规卷积的输出。所有层级层的连接提供了多级特征并保留了多样性，因此模型可以在广阔的特征空间中搜索理想结果。

3) 实验中联合使用了 GCB-net 和 BLS。BLS 被用于增强 GCB-net 的特征。在脑电图情绪识别任务中，与其它模型相比，GCB-net 和 BLS 取得了更优的结果，展示了 GCB-net 和 BLS 在脑电图情绪识别中的鲁棒分类能力。

2 初步

本节介绍了常规卷积、图卷积和广义学习系统的基本理论，它们是构建 GCB-net 的基础知识。

2.1 卷积操作

卷积是一个涉及两个函数（数据函数和核函数）的操作，通常表示为算子 。大多数神经网络实现卷积时不进行核翻转，一维卷积操作可以简单地定义为

$$S(i) = (X * W)(i) = \sum_n X(i+n)W(n); \quad (1)$$

其中 $X(\cdot)$ 是一维数据，$W(\cdot)$ 是一维核函数。由此推广，更高维的卷积可以以类似的方式表示，例如二维卷积操作

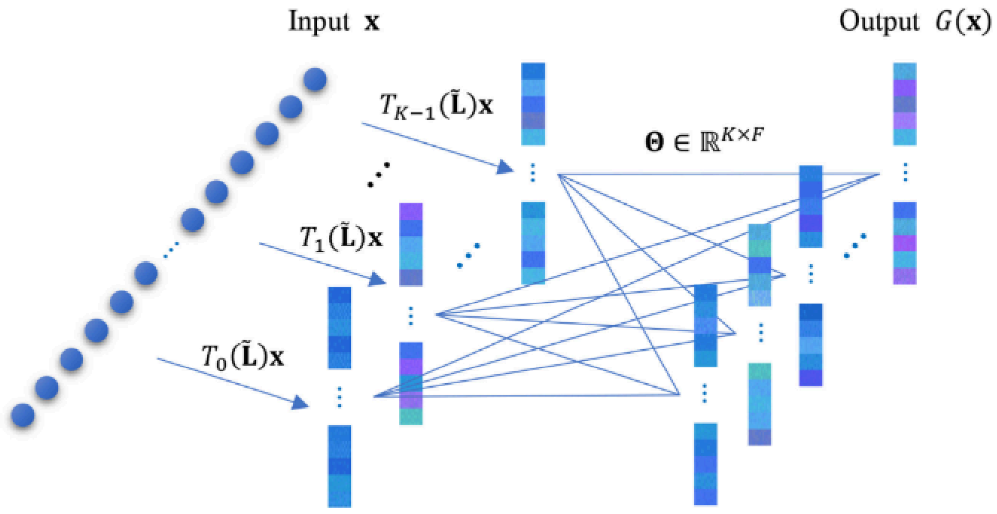

Fig. 1. The framework of graph convolution. The first layer is input data, the second layer is the Chebyshev approximation of graph transformation and the third layer is the result of graph filtering.

$$S(i,j) = (X * W)(i,j) = \sum_m \sum_n X(i+m, j+n)W(m,n). \tag{2}$$

Generally, the kernel is much smaller than input data to exploit local properties. In deep networks, usually stacking multiple layers to obtain a larger responsive region of the input and create higher-level representations.

2.2 Graph Convolutional Neural Networks

A undirected or connected graph can be represented as $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{A})$, where $\mathcal{V}$ is a vertex set with $|\mathcal{V}| = n$ nodes and $\mathcal{E}$ is a finite set of edges $(v_i \in \mathcal{V}, v_j \in \mathcal{V})$. $\mathbf{A} \in \mathbb{R}^{n\times n}$ is a binary or weighted adjacency matrix describing connections between nodes. The graph Laplacian matrix of $\mathcal{G}$ is defined as $\mathbf{L} = \mathbf{D} - \mathbf{A}$, where $\mathbf{D} \in \mathbb{R}^{n\times n}$ is the degree matrix with diagonal entries $D_{ii} = \sum_j A_{ij}$. The normalized version can be written as $\mathbf{L} = \mathbf{I}_n - \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}$ and $\mathbf{I}_n$ is a identify matrix. The eigendecomposition of $\mathbf{L}$ is $\mathbf{U}\Lambda\mathbf{U}^\top$, where $\mathbf{U} = [\mathbf{u}_0, \mathbf{u}_1, \ldots, \mathbf{u}_{n-1}] \in \mathbb{R}^{n\times n}$ is a square matrix of eigenvectors ordered by eigenvalues and $\mathbf{\Lambda} = diag(\lambda_0, \lambda_1, \ldots, \lambda_{n-1}) \in \mathbb{R}^{n\times n}$ is a diagonal matrix with corresponding eigenvalues.

Bruna et al. [33] used $\mathbf{U}$ as Fourier basis. Then graph Fourier transform is written as

$$\hat{\mathbf{x}} = \mathbf{U}^\top \mathbf{x}, \tag{3}$$

and its inverse transform is computed as

$$\mathbf{x} = \mathbf{U}\hat{\mathbf{x}}. \tag{4}$$

In the spectral domain, the convolution operation of two signals on graph $*_\mathcal{G}$ can be defined as

$$\mathbf{x}_1 *_\mathcal{G} \mathbf{x}_2 = \mathbf{U}\big((\mathbf{U}^\top\mathbf{x}_1) \odot (\mathbf{U}^\top\mathbf{x}_2)\big), \tag{5}$$

where $\odot$ represents element-wise Hadamard product. And dealing a signal $\mathbf{x}$ with filter g_θ can be formulated as follows:

$$g_\theta *_\mathcal{G} \mathbf{x} = g_\theta(\mathbf{L})\mathbf{x} = g_\theta(\mathbf{U}\Lambda\mathbf{U}^\top)\mathbf{x} = \mathbf{U}g_\theta(\mathbf{\Lambda})\mathbf{U}^\top\mathbf{x}. \tag{6}$$

Directly using (6) is computationally expensive. First, computing full eigenvectors of the Laplacian matrix is expensive for large graphs. Second, multiplication with the Fourier basis $\mathbf{U}$ requires $\mathcal{O}(n^2)$ operations. To circumvent problems, Defferrard et al. [34] suggested spectral filter $g_\theta(\cdot)$ can be approximated by a truncated Chebyshev expansion

$$g'_\theta(\mathbf{\Lambda}) = \sum_{k=0}^{K-1} \theta_k T_k(\tilde{\mathbf{\Lambda}}), \tag{7}$$

where K is the number of Chebyshev polynomial parameters, and $\theta_0, \theta_1, \ldots, \theta_{K-1}$ are Chebyshev coefficients. $\tilde{\mathbf{\Lambda}} = 2\mathbf{\Lambda}/\lambda_{max} - \mathbf{I}_n$ is a diagonal matrix of rescaled eigenvalues. The Chebyshev polynomial $T_k(\cdot)$ can be recursively computed as $T_k(x) = 2xT_{k-1}(x) - T_{k-2}(x)$, where $T_0(x) = 1$ and $T_1(x) = x$. Then graph filtering operation can be approximately written as

$$\begin{aligned} g'_\theta *_\mathcal{G} \mathbf{x} &= g'_\theta(\mathbf{L})\mathbf{x} \\ &= \sum_{k=0}^{K-1} \theta_k T_k(\tilde{\mathbf{L}})\mathbf{x} \\ &= \sum_{k=0}^{K-1} \theta_k \bar{\mathbf{x}}_k, \end{aligned} \tag{8}$$

where $\bar{\mathbf{x}}_k = T_k(\tilde{\mathbf{L}})\mathbf{x} \in \mathbb{R}^n$ and it can be recursively computed as $\bar{\mathbf{x}}_k = 2\tilde{\mathbf{L}}\bar{\mathbf{x}}_{k-1} - \bar{\mathbf{x}}_{k-2}$ with $\bar{\mathbf{x}}_0 = \mathbf{x}$ and $\bar{\mathbf{x}}_1 = \tilde{\mathbf{L}}\mathbf{x}$. Similarly, $\tilde{\mathbf{L}} = 2\mathbf{L}/\lambda_{max} - \mathbf{I}_n$ is a rescaled Laplacian matrix. Therefore, the operation of graph convolution can be represented as $g'_\theta(\mathbf{L})\mathbf{x} = [\bar{\mathbf{x}}_0, \bar{\mathbf{x}}_1, \ldots, \bar{\mathbf{x}}_{K-1}]\boldsymbol{\theta}$. And the time complexity is reduced to $\mathcal{O}(K|\mathcal{E}|)$. The framework of graph convolution is shown in Fig. 1.

Song et al. [36] proposed DGCNN to learn the adjacency matrix $\mathbf{A}$ of graph through back propagation (BP) method. The general loss function with cross entropy is

$$Loss = cross\_entropy(\mathbf{y}, \mathbf{y}') + \alpha\|\mathbf{\Theta}\|, \tag{9}$$

where $\mathbf{y}$ and $\mathbf{y}'$ are vectors of ture labels and predicted labels respectively. $\mathbf{\Theta}$ is a matrix of all parameters learned in the DGCNN model and α is a regularization coefficient. The partial derivative of the loss function w.r.t. adjacency matrix $\mathbf{A}$ can be deduced as

$$\frac{\partial Loss}{\partial \mathbf{A}} = \frac{\partial cross\_entropy(\mathbf{y}, \mathbf{y}')}{\partial \tilde{\mathbf{L}}} \cdot \frac{\partial \tilde{\mathbf{L}}}{\partial \mathbf{A}} + \alpha \frac{\partial \|\mathbf{\Theta}\|}{\partial \mathbf{A}}. \tag{10}$$

And the model can update $\mathbf{A}$ by the following formula:

$$\mathbf{A} = (1-\rho)\mathbf{A} + \rho\frac{\partial Loss}{\partial \mathbf{A}}, \tag{11}$$

where ρ is the learning rate of the model.

2.3 Broad Learning System

BLS [39] offers an alternative learning structure that expand features to broad space through random methods. The network is mainly constructed with feature nodes and enhancement nodes. Feature nodes are randomly mapped from input and enhancement nodes are also randomly expanded from feature nodes. Then the output of BLS is solved by joint of feature nodes and enhancement nodes. Notably, the parameters of feature nodes and enhancement nodes are generated once and only the weights connected to the output layer need to be learned in training procedure. Hence, BLS can be modeled in a fast way, especially compared with other deep architectures.

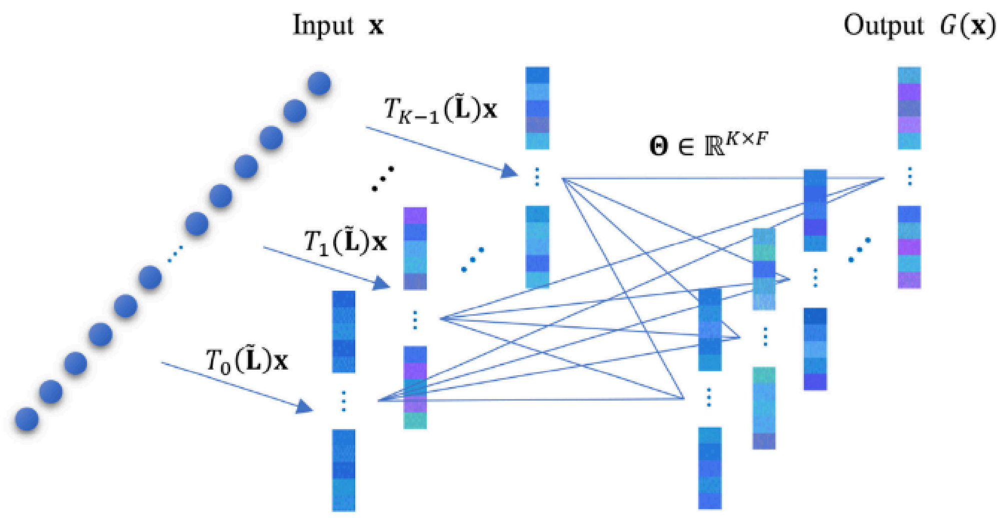

图 1. 图卷积框架。第一层是输入数据，第二层是图转换的切比雪夫近似，第三层是图过滤的结果。

Sði; jÞ ¼ ðX W Þði; jÞ
X X
¼ X i þ m; j þ nð ÞW m; nðÞ: (2)
m n

通常，核的大小远小于输入数据，以便利用局部特性。在深度网络中，通常堆叠多层以获得更大的输入响应区域，并创建更高级别的表示。

2.2 图卷积神经网络

一个无向或连通图可以表示为 G = (V, E, A)，其中 V 是一个包含 Vj j = n 个节点的顶点集，E 是一个有限边集 v ∈ V； v ∈ V
. A 2 是一个描述节点之间连接的二进制或加权邻接矩阵。G 的图拉普拉斯矩阵定义为 L = D - A，其中 D ∈ R 是度矩阵，其对角线元素为 D = ∑j A。归一化版本可以写成 L = I - DAD，其中 I 是单位矩阵。L 的特征分解为 ULU，其中 U = [u₁, u₂, ..., uₙ] ∈ R 是按特征值排序的特征向量矩阵，且 L = diag(λ₁, λ₂, ..., λₙ)。
Þ 2 是一个对角矩阵，具有相应的特征值。
Bruna 等人[33]使用 U 作为傅里叶基。然后图傅里叶变换表示为

x̂ = Ux; (3)

其逆变换计算为

x = U^x: (4)

在频谱域中，图上两个信号的卷积操作可以定义为

x x¼ U Ux Ux ; (5)

其中 表示逐元素 Hadamard 乘积。对信号 x 使用滤波器 g 可以表示为如下公式：

g ¼ gðLÞg ¼ gðULUÞg ¼ UgðLÞUg: (6)

直接使用公式(6)计算成本很高。首先，对于大型图来说，计算拉普拉斯矩阵的完整特征向量很昂贵。其次，与傅里叶基 U 相乘需要 O(n) 次操作。为了解决这个问题，Defferrard 等人

[34]建议频谱滤波器 g 可以近似为截断切比雪夫展开

K1
X
gðLÞ ¼ uTð ~LÞ; (7)
k¼0

其中 K 是切比雪夫多项式参数的数量，而 u₁, u₂, ... , uₙ是切比雪夫系数。
L̃ ¼ 2L= Iis a diagonal matrix of rescaled eigenvalues. The Chebyshev polynomial TðÞ can be recursively computed as T ðxÞ ¼ 2xTðxÞ T ðxÞ, where TðxÞ ¼ 1 and TðxÞ ¼ x. Then graph filtering operation can be approximately written as

gx ¼ gðLÞx
K1
X
¼ uTð~LÞx (8)
k¼0
K1
X
¼ ux;
k¼0

在 x¼ Tð~LÞx 属于 Rand 它可以递归地计算为 x¼ 2~L x - x，其中 x ¼ x 和 x ¼ ~L x 。类似地，
L̃ ¼ 2L/ - I 是一个重新缩放的拉普拉斯矩阵。因此，图卷积的操作可以表示为 gðLÞx ¼ x ; x ; . . . ; x ½ Šuu 。时间复杂度降低到 OðK Ej jÞ 。图卷积的框架如图 1 所示。
Song 等人 [36] 提出了 DGCNN，通过反向传播 (BP) 方法学习图的邻接矩阵 A 。具有交叉熵的通用损失函数为

Loss ¼ 交叉熵ðy; yÞ + a Qk k (9)

y 和 y 分别是真实标签和预测标签的向量。Q 是 DGCNN 模型中学习到的所有参数的矩阵，a 是正则化系数。损失函数相对于邻接矩阵 A 的偏导数可以推导为

@Loss / @A ¼ @交叉熵(y; yÞ / @~L @~Lþ a @Qk k / @A @A (10)

模型可以通过以下公式更新 A：

A ¼ ð1 rÞA þ r @Loss / @A; (11)

其中 r 是模型的学习率。

2.3 广义学习系统

BLS [39] 提供了一种替代的学习结构，通过随机方法扩展特征以扩展到广阔的空间。该网络主要由特征节点和增强节点构成。特征节点从输入中随机映射，增强节点也从特征节点中随机扩展。然后通过特征节点和增强节点的联合来求解 BLS 的输出。值得注意的是，特征节点和增强节点的参数一次性生成，在训练过程中只需要学习与输出层相连的权重。因此，BLS 可以快速建模，尤其与其他深度架构相比。

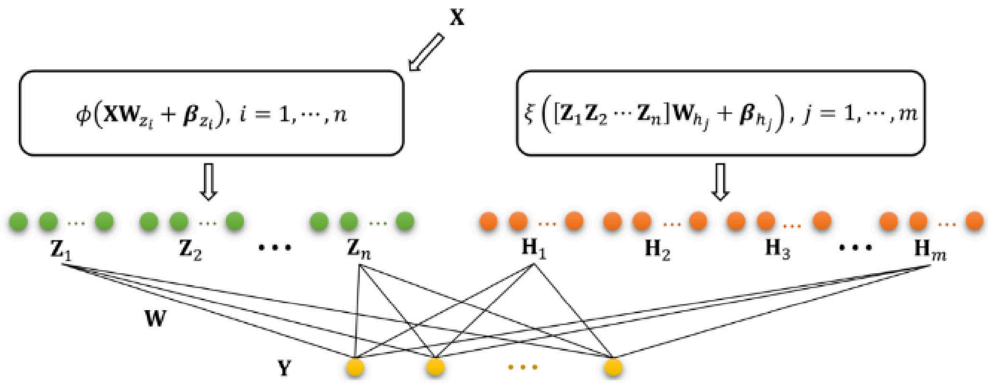

Fig. 2. The framework of Broad Learning System (BLS). The green circles represent the feature nodes, orange circles represent enhancement nodes and yellow circles represent predicted results.

Denoting input data set and label matrix as $\mathbf{X} \in \mathbb{R}^{N\times M}$ and $\mathbf{Y} \in \mathbb{R}^{N\times C}$ respectively. N is the number of samples, M is the dimension of features and C is the number of classes. Feature nodes can be generated as

$$\mathbf{Z}_i \triangleq \phi(\mathbf{X}\mathbf{W}_{z_i} + \boldsymbol{\beta}_{z_i}),\ i = 1, 2, \ldots, n, \tag{12}$$

where weighting matrix $\mathbf{W}_{z_i} \in \mathbb{R}^{M\times N_f}$ and bias matrix $\boldsymbol{\beta}_{z_i} \in \mathbb{R}^{1\times N_f}$ are random generated. N_f is the dimension of generated features and $\phi(\cdot)$ is activation function. The collection of all feature nodes is denoted as $\mathbf{Z}^n \triangleq [\mathbf{Z}_1, \ldots, \mathbf{Z}_n] \in \mathbb{R}^{N\times(n*N_f)}$, where n is the number of groups of feature nodes. And similarly the definition of enhancement nodes can be written in following formula:

$$\mathbf{H}_j \triangleq \xi(\mathbf{Z}^n\mathbf{W}_{h_j} + \boldsymbol{\beta}_{h_j})\ j = 1, 2, \ldots, m, \tag{13}$$

where parameter matrixs $\mathbf{W}_{h_j} \in \mathbb{R}^{(n*N_f)\times N_e}$ and $\boldsymbol{\beta}_{h_j} \in \mathbb{R}^{1\times N_e}$ are also randomly initialized. N_e is the dimension of enhancement features and $\xi(\cdot)$ is a nonlinear activation function. The collection of all enhancement nodes is denoted as $\mathbf{H}^m \triangleq [\mathbf{H}_1, \ldots, \mathbf{H}_m] \in \mathbb{R}^{N\times(m*N_e)}$, where m is the number of groups of enhancement nodes and usually m is much larger than n.

Integrating feature nodes and enhancement nodes together, final output can be computed as

$$\begin{aligned}\mathbf{Y} &= [\mathbf{Z}_1, \ldots, \mathbf{Z}_n|\mathbf{H}_1, \ldots, \mathbf{H}_m]\mathbf{W}\\ &= [\mathbf{Z}^n|\mathbf{H}^m]\mathbf{W}.\end{aligned} \tag{14}$$

The connecting weights $\mathbf{W} \in \mathbb{R}^{(n*N_f+m*N_e)\times C}$ can be solved by pseudo-inverse method and the analytical solution can be written as $\mathbf{W} \triangleq [\mathbf{Z}^n|\mathbf{H}^m]^+\mathbf{Y}$. The framework of BLS is illustrated in Fig. 2.

3 Graph Convolutional Broad Network

This section introduces the modeling way of GCB-net and its improved algorithm in detail, and gives the illustrations of framework and algorithm tables.

3.1 Graph Convolutional Broad Model

Graph Convolutional Broad Network (GCB-net) is designed for discovering the deeper hidden information of graph-structured data. The framework of GCB-net for classification is illustrated in Fig. 3. The network first applys Chebyshev graph convolution to deal with the irregular grid data. Then the regular convolution are utilized to extract higher-level features. After all convolutions, outputs of different layers are flatten to one-dimension vectors and concatenated together. Then, the network is followed by fully connected layer and predicted with softmax function.

Defining the matrix of input graph-structured data as $\mathbf{x} \in \mathbb{R}^{M\times D_1\times\cdots\times D_h}$ and vector of output labels as $\mathbf{y} \in \mathbb{R}^{C\times 1}$, where M is the number of vertexes in graph, h is the dimension of each node and C is the number of data categories. The output of $K-1$ order Chebyshev graph convolution can be represented as

$$G(\mathbf{x}) = \mathrm{ReLU}\big(\big[T_0(\tilde{\mathbf{L}})\mathbf{x}, \ldots, T_{K-1}(\tilde{\mathbf{L}})\mathbf{x}\big]\boldsymbol{\Theta}\big), \tag{15}$$

where $\boldsymbol{\Theta} \in \mathbb{R}^{K\times F}$ is the matrix formed by coefficients of Chebyshev filters and F is the number of filters. The active function ReLU $(x) = \max(x, 0)$ is used to increase nonlinearity. Then graph convolutional output $G(\mathbf{x})$ is fed into the following regular convolutional layers. After a few layers of regular convolution, outputs of all layers are linked together. The concatenating features of graph convolution and regular convolution can be defined as follows:

$$GCB(\mathbf{x}) = [G, C1, C2\ldots, Cl], \tag{16}$$

where $G = \mathrm{flat}(G(\mathbf{x}))$ represents the flattening vector of graph convolutional output, $C1 = \mathrm{flat}(C_1(\mathbf{x}))$ represents the flattening vector of the first regular convolutional output. $C_1(\mathbf{x}) = \mathrm{conv}(G(\mathbf{x}), h)$, where $h = 1, 2, 3$ is the dimension of convolutional kernel and it depends on the type of data, $\mathrm{conv}(\cdot, h)$ denotes h-D convolutional operation. l is the number of regular convolutional layers. By analogy, $Ck = \mathrm{flat}(C_k(\mathbf{x})), k = 1, 2, \ldots, l$ represents the flattening vector of kth h-D convolutional output and $C_k(\mathbf{x}) = \mathrm{conv}(C_{k-1}(\mathbf{x}), h)$. Then the fully connected layer can be computed by

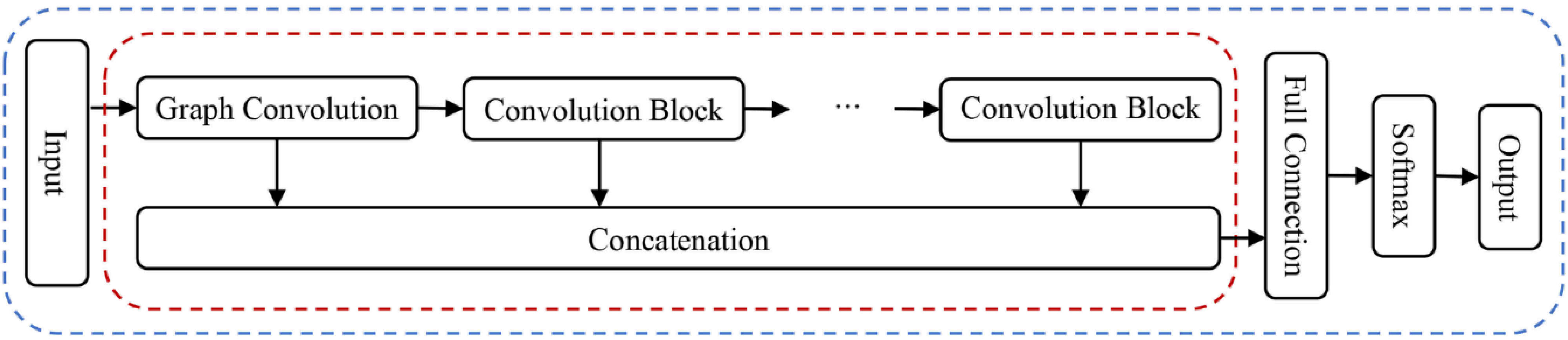

Fig. 3. The framework of GCB-net. The input data first goes through graph convolution and then feeds into series of regular convolution. The concatenation of these convolution is linked to a fully connected layer and finally outputs with softmax fuction. The content in red dotted box represents the feature construction of GCB-net, and the content in blue dotted box represents a complete process of GCB-net.

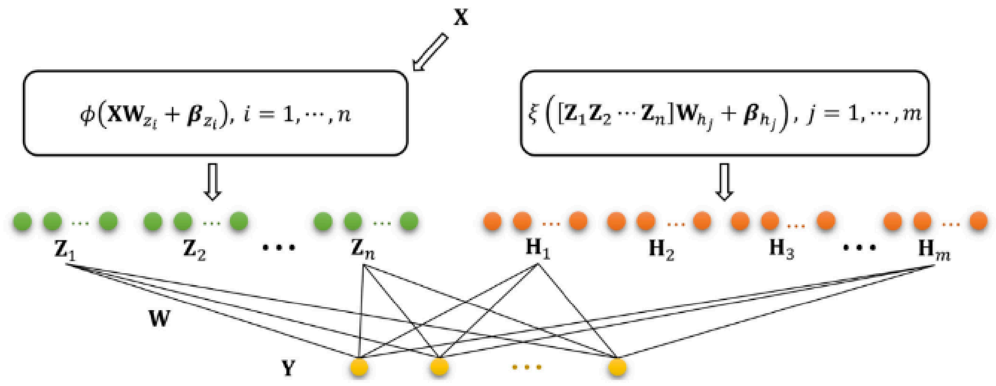

图 2. 广义学习系统（BLS）的框架。绿色圆圈表示特征节点，橙色圆圈表示增强节点，黄色圆圈表示预测结果。

将输入数据集和标签矩阵分别表示为 X ∈ R 和 Y ∈ R。N 是样本数量，M 是特征维度，C 是类别数量。特征节点可以生成为

Z, fðXWþ bbÞ; i ¼ 1; 2; . . . ; n; (12)

权重矩阵 W2 和偏置矩阵

b b2 稀疏随机生成。N 是生成特征的维度，fðÞ是激活函数。所有特征节点的集合表示为 Z，Z；...；Z½ Š 2 R，其中 n 是特征节点组的数量。类似地，增强节点的定义可以写成以下公式：

H, ƯZWþ bbÞ j ¼ 1; 2; . . . ; m; (13)

其中参数矩阵 W2 Rand bb2 R 也被随机初始化。N 是增强特征的维度， Ư åÞ 是一个非线性激活函数。所有增强节点的集合表示为 H，H；. . . ；H½ Š 2 R，其中 m 是增强节点组的数量，通常 m 远大于 n。

将特征节点和增强节点结合起来，最终输出可以计算为

Y ¼ Z; . . . ; ZjH; . . . ; H½ W
ZjH½ W: (14)

连接权重 W ∈ R 可以通过伪逆方法求解，解析解可以表示为 W = ZjH½ Y。BLS 框架如图 2 所示。

3 图卷积广网络

本节详细介绍了 GCB-net 的建模方式及其改进算法，并给出了框架和算法表的示意图。

3.1 图卷积广模型

图卷积广网络（GCB-net）旨在发现图结构数据的更深层次隐藏信息。GCB-net 用于分类的框架如图 3 所示。该网络首先应用切比雪夫图卷积处理不规则网格数据。然后利用常规卷积提取高级特征。经过所有卷积后，不同层的输出被展平为一维向量并连接在一起。接着，网络通过全连接层，并使用 softmax 函数进行预测。

将输入图结构数据矩阵定义为 x ∈ R^M，将输出标签向量定义为 y ∈ R，其中 M 是图中的顶点数，h 是每个节点的维度，C 是数据类别数。K 阶切比雪夫图卷积的输出可以表示为

G(x) = ReLU(T(~L)x; ..., T(~L)x) Q ; (15)

其中 Q ∈ R 是切比雪夫滤波器系数构成的矩阵，F 是滤波器数量。使用激活函数 ReLU(x) = max(x; 0)来增加非线性。然后将图卷积输出 G(x)输入到后续的常规卷积层中。经过几层常规卷积后，所有层的输出被连接在一起。图卷积和常规卷积的拼接特征可以定义为如下：

GCB(x) = G; C1; C2; ..., Cl Š; (16)

其中 G ¼ flatðGðxÞÞ 表示图卷积输出的展平向量，C1 ¼ flatðCðxÞÞ 表示第一个常规卷积输出的展平向量。CðxÞ ¼ convðGðxÞ; hÞ，其中 h ¼ 1; 2; 3 是卷积核的维度，它取决于数据的类型，convð; hÞ 表示 h 维卷积操作。l 是常规卷积层的数量。类似地，Ck ¼ flatðCðxÞÞ; k ¼ 1; 2; . . . ; l 表示第 k 个 h 维卷积输出的展平向量，且 C ðxÞ ¼ convðCðxÞ; hÞ。然后可以通过全连接层进行计算：

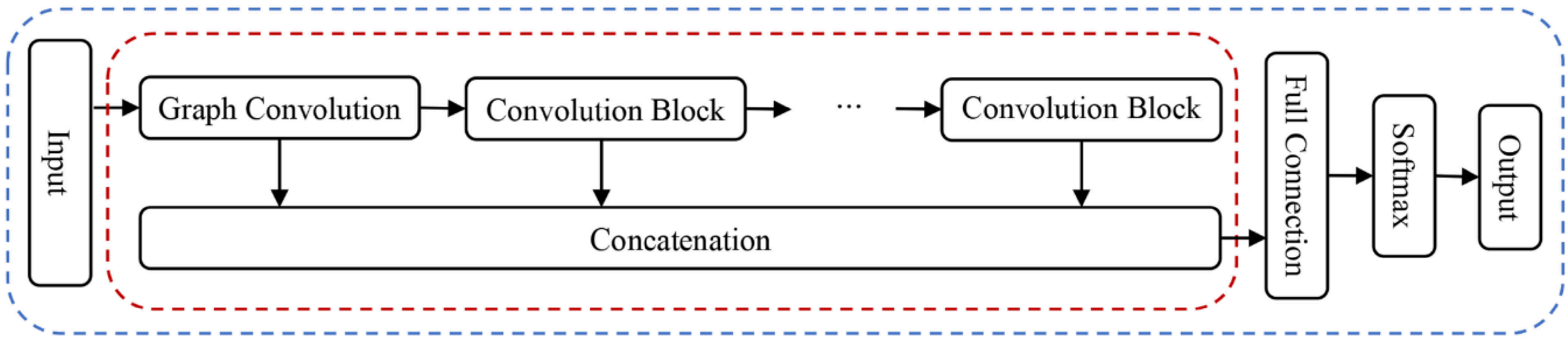

图 3. GCB-net 的框架。输入数据首先经过图卷积，然后输入到一系列常规卷积中。这些卷积的连接链接到一个全连接层，并最终通过 softmax 函数输出。红色虚线框中的内容表示 GCB-net 的特征构建，蓝色虚线框中的内容表示 GCB-net 的完整过程。

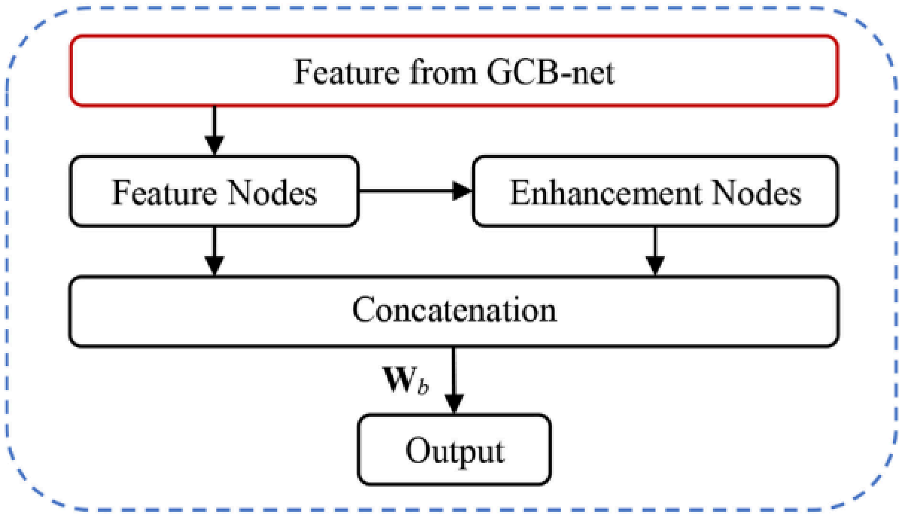

Fig. 4. The diagram of enhancing GCB-net with BLS. Content in red box is the concatenating features before full connection in GCB-net. Feature nodes are randomly mapped by input features and enhancement nodes are randomly mapped by feature nodes. Finally linked all nodes together to compute output with weighting matrix $\mathbf{W}_b$.

$$FC(\mathbf{x}) = GCB(\mathbf{x})\mathbf{W}_f, \tag{17}$$

where $\mathbf{W}_f \in \mathbb{R}^{N_{gcb} \times C}$ is the affine matrix, where N_{gcb} is the feature number of $GCB(\mathbf{x})$. The final predicted layer can be normalized by softmax function

$$\mathbf{y} = \mathrm{softmax}(FC(\mathbf{x})), \tag{18}$$

and where

$$\mathrm{softmax}(c, x_1, x_2, \ldots, x_C) = \frac{e^{x_c}}{\sum_{i=1}^{C} e^{x_i}}, \tag{19}$$

ensures that results are transformed to a probability distribution. The parameters of GCB-net can also be learned by BP method. The detailed implementation steps of GCB-net are summarized in Algorithm 1.

Algorithm 1. The Implementation Steps of Original GCB-Net for Classification

Input: the graph-structured data $\mathbf{x}$, the number of Chebyshev coefficients K, the layers of regular convolution l, the dimension of convolutional kernel h, the number of label class C;
Output: the label vector $\mathbf{y}_s$;
Calculate the graph convolutional layer $G(\mathbf{x})$ via (15);
Calculate the first regular convolutional layer $C_1(\mathbf{x}) = conv(G(\mathbf{x}), h)$;
for $k = 2, \ldots, l$ **do**
  Calculate the kth regular convolutional layer $C_k(\mathbf{x}) = conv(C_{k-1}(\mathbf{x}), h)$;
Flatten the outputs of graph convolution $G(\mathbf{x})$ to one-dimensional vectors G;
for $k = 1, \ldots, l$ **do**
  Flatten the outputs of regular convolution $C_k(\mathbf{x})$ to one-dimensional vectors Ck;
Concatenate the flatting vectors of all layers together to get $GCB(\mathbf{x}) = [G, C1, \ldots, Cl]$;
Calculate the fully connected layer $FC(\mathbf{x}) = GCB(\mathbf{x})\mathbf{W}_f$,
Calculate the predicted matrix $\mathbf{y}_s = \mathrm{softmax}(FC(\mathbf{x}))$;

3.2 Improved Algorithm with BLS

To improve the performance of GCB-net, BLS can be used to enhance feature from GCB-net. First, GCB feature are expanded to random broad spaces through feature mapping and enhancement mapping, then output can be predicted by connecting of feature nodes and enhancement nodes. The diagram of enhancing GCB-net with BLS is drawn in Fig. 4. Specifically, the active function $\phi(\cdot)$ of feature nodes is chosen as the identify function and feature nodes of GCB-net can be written as

$$\mathbf{Z}_i \triangleq GCB(\mathbf{X})\mathbf{W}_{z_i} + \boldsymbol{\beta}_{z_i}, \; i = 1, 2, \ldots, n, \tag{20}$$

The nonlinear function $\xi(\cdot)$ of enhancement nodes is tangent sigmoid function

$$\mathrm{tansig}(x) = \frac{2}{1 + e^{2x}} - 1, \tag{21}$$

and enhancement nodes in the model can be computed as

$$\mathbf{H}_j \triangleq \mathrm{tansig}(\mathbf{Z}^n \mathbf{W}_{h_j} + \boldsymbol{\beta}_{h_j}) \; j = 1, 2, \ldots, m. \tag{22}$$

To prevent the over-fitting problem, regular l_2 norm regularization is added to the loss function, which can be written as follows:

$$\underset{\mathbf{W}_b}{\arg\min} : \|[\mathbf{Z}^n|\mathbf{H}^m]\mathbf{W}_b - \mathbf{Y}\|_2^2 + \lambda\|\mathbf{W}_b\|_2^2, \tag{23}$$

where $\mathbf{W}_b$ is the weighting matrix and λ is the regularization coefficient. Taking the derivative of (23) w.r.t $\mathbf{W}_b$ and setting it to zero, then the connecting weights $\mathbf{W}_b$ can be easily computed as

$$\mathbf{W}_b = \left([\mathbf{Z}^n|\mathbf{H}^m]^\top [\mathbf{Z}^n|\mathbf{H}^m] + \lambda \mathbf{I}\right)^{-1} [\mathbf{Z}^n|\mathbf{H}^m]^\top \mathbf{Y}. \tag{24}$$

The detail of implementing GCB-net with BLS for classification problems is summarized in Algorithm 2.

Algorithm 2. The Implementation Steps of GCB-Net with BLS for Classification

Input: the feature from GCB-net $GCB(\mathbf{X})$, the number of groups of feature nodes n, the number of groups of enhancement nodes m, the dimension of generated features N_f, the dimension of enhancement features N_e and the regularization coefficient λ;
Output: the label matrix $\mathbf{Y}_b$;
for $i = 1, 2, \ldots, n$ **do**
  Randomly initialize $\mathbf{W}_{z_i}$ and $\boldsymbol{\beta}_{z_i}$;
  Calculate $\mathbf{Z}_i \triangleq GCB(\mathbf{X})\mathbf{W}_{z_i} + \boldsymbol{\beta}_{z_i}$;
Collect all feature nodes $\mathbf{Z}^n \triangleq [\mathbf{Z}_1, \ldots, \mathbf{Z}_n]$;
for $j = 1, 2, \ldots, m$ **do**
  Randomly initialize $\mathbf{W}_{h_j}$ and $\boldsymbol{\beta}_{h_j}$;
  Calculate $\mathbf{H}_j \triangleq \mathrm{tansig}(\mathbf{Z}^n\mathbf{W}_{h_j} + \boldsymbol{\beta}_{h_j})$;
Collect all enhancement nodes $\mathbf{H}^m \triangleq [\mathbf{H}_1, \ldots, \mathbf{H}_m]$;
Calculate the connecting weights $\mathbf{W}_b$ via (24);
Calculate the label matrix $\mathbf{Y}_b = [\mathbf{Z}^n|\mathbf{H}^m]\mathbf{W}_b$;

4 Experiments and Analysis for EEG Emotion Recognition

The experiments were conducted on EEG emotion data, and this section contains descriptions of dataset, architecture of GCB-net in EEG emotion recognition, experimental setting, results and analysis.

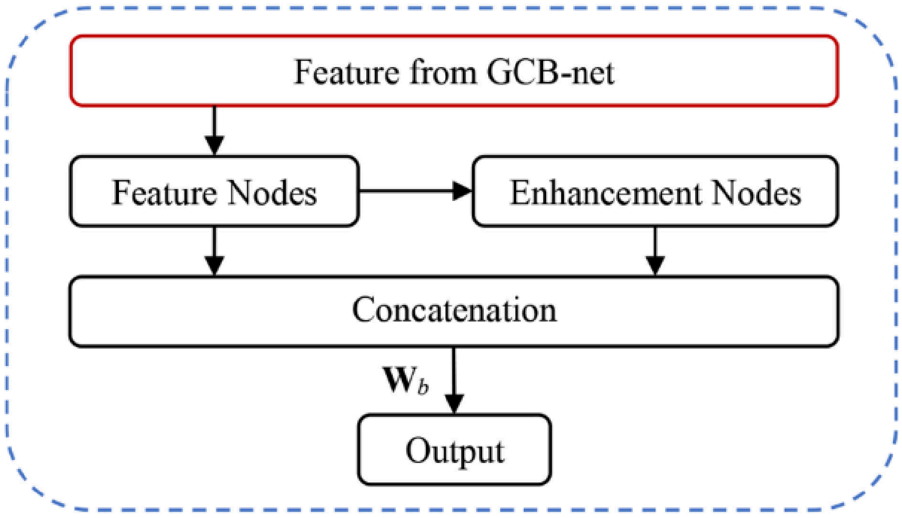

图 4. 使用 BLS 增强 GCB-net 的示意图。红色框内是 GCB-net 全连接前的特征拼接。特征节点由输入特征随机映射，增强节点由特征节点随机映射。最后将所有节点连接起来，使用权重矩阵 W 计算输出。

FCðxÞ ¼ GCB xð ÞW; (17)

W2 是仿射矩阵，其中 N 是 GCB xð Þ 的特征数量。最终预测的层可以通过 softmax 函数进行归一化

y ¼ softmaxðFCðxÞÞ; (18)

并且

softmaxðc; x; x; . . . ; x Þ ¼ P e / i¼1 e ; (19)

确保结果转换为概率分布。GCB-net 的参数也可以通过 BP 方法学习。GCB-net 的详细实现步骤总结在算法 1 中。

算法 1. 原始 GCB-Net 分类的实现步骤 输入：图结构数据 x，切比雪夫系数的数量 K，常规卷积的层数 l，卷积核的维度 h，标签类的数量 C；

输出：标签向量 y； 1: 通过(15)计算图卷积层 G(x)； 2: 计算第一层常规卷积层 C(x) =

convðGðxÞ; hÞ;
3: for k ¼ 2; . . . ; l do
4: 计算第 k 个常规卷积层 CðxÞ ¼
convðC ðxÞ; hÞ;
5: 将图卷积 GðxÞ 的输出展平为一维向量 G； 6: 对于 k = 1; . . . ; l 执行
7: 将常规卷积 C ðxÞ 的输出展平为一维向量 Ck； 8: 将所有层的展平向量连接在一起得到

GCBðxÞ = [G; C1; . . . ; Cl];
9: 计算全连接层 FCðxÞ = GCB xð ÞW， 10: 计算预测矩阵 y = softmaxðFCðxÞÞ;

3.2 带有 BLS 的改进算法

为了提升 GCB-net 的性能，可以使用 BLS 来增强 GCB-net 的特征。首先，通过特征映射和增强映射，将 GCB 特征扩展到随机广阔的空间中

然后输出可以通过连接特征节点和增强节点来预测。增强 GCB-net 与 BLS 的示意图绘制在图 4 中。具体来说，特征节点的激活函数 fðÞ 被选择为恒等函数，GCB-net 的特征节点可以表示为

Z, GCBðXÞWþ bb; i ¼ 1; 2; . . . ; n; (20)

增强节点的非线性函数ðÞ是切比雪夫 sigmoid 函数

tansigðxÞ ¼ 2 / 1 þ e 1; (21)

并且模型中的增强节点可以计算为

H, tansigðZWþ bbÞ j ¼ 1; 2; . . . ; m: (22)

为防止过拟合问题，在损失函数中添加了正则 lnorm 正则项，其表达式如下：

arg min : ZjH½ ŠW Y kþ Wk k; (23)
Wb

其中 W 是权重矩阵， 是正则化系数。对(23)关于 W 求导并设其为零，则连接权重 W 可以很容易地计算出来。

W¼ ZjH½ ŠZjH½ Š þ I ZjH Š Y: (24)

将 GCB-net 与 BLS 结合用于分类问题的实现细节总结在算法 2 中。

算法 2. GCB-Net 与 BLS 结合用于分类的实现步骤 输入：来自 GCB-net GCB(X)的特征，特征节点组的数量 n，增强节点组的数量 m，生成特征的维度 N，增强特征的维度 N 以及正则化系数λ；

输出：标签矩阵 Y；
1: for i ¼ 1; 2; . . . ; n do
2: 随机初始化 W 和 bb;
计算 Z, GCB(X)W+bb; 4: 收集所有特征节点 Z, Z; . . . ; Z½ Š;
5: for j ¼ 1; 2; . . . ; m do 6: 随机初始化 W 和 bb; 7: 计算 H, tansig(ZW+bbÞ; 8: 收集所有增强节点 H, H; . . . ; H½ Š; 9: 计算连接权重 Wvia (24); 10: 计算标签矩阵 Y¼ ZjH½ ŠW;

脑电图情绪识别的实验与分析 脑电图情绪数据的实验在此进行，本节包含数据集、脑电图情绪识别中 GCB-net 架构、实验设置、结果和分析的描述。

TABLE 1
Summary Table of SEED

Character	Details				
Feature	DE	PSD	DASM	RASM	DCAU
Used electrodes	62	62	27 pairs	27 pairs	23 pairs
Frequency band	$\delta, \theta, \alpha, \beta, \gamma$ bands				
Rating metric	posive, neutral, negative				

TABLE 2
Summary Table of DREAMER Dataset

Character	Details
Feature	PSD
Used electrodes	14
Frequency band	θ, α, β bands
Rating metric	Arousal, Valence, Dominance
Rating scale	$1, 2, 3, 4, 5$

4.1 Dataset

We evaluated our method on SJTU emotion EEG dataset (SEED) [44] and DREAMER dataset [45].

4.1.1 SEED

There were 15 participants (7 males and 8 females) in SEED data acquisition. All participants watched 15 movie clips containing 5 positive emotions, 5 neutral emotions and 5 negative emotions. Everyone participated in experiments 3 times at intervals of a week or longer.

The EEG data was recorded from a 62-channel electrode cap and down-sampled to a 200 Hz sampling rate. The data releaser removed the noise and artifacts with a band-pass filter between 0.3 to 50 Hz. Then, the linear dynamic system (LDS) was applied for further smoothing. SEED provides several extracted EEG features, containing differential entropy (DE), power spectral density (PSD), differential asymmetry (DASM), rational asymmetry (RASM) and differential caudality (DCAU). DE feature and PSD feature extract contents about frequency and energy spectrum, DASM feature and RASM feature obtain asymmetrical information of EEG channels, and DCAU feature computes the differences between channel pairs. Each kind of features contains five frequency bands: δ band (1-3 Hz), θ band (4-7 Hz), α band (8-13 Hz), β band (14-30 Hz) and γ band (31-50 Hz). The characters of SEED are listed in Table 1.

4.1.2 DREAMER Dataset

In DREAMER dataset, EEG data were acquired by 23 participants (14 males and 9 females). 18 film clips were selected to evoke 9 emotions: amusement, excitement, happiness, calmness, anger, disgust, fear, sadness and surprise. The duration of films are from 65 to 393 s, and only the last 60 s of EEG signals elicited by films were used. After watching the film, Participants assessed self emotion form valence (negative to positive), arousal (level of emotional activation), and dominance (strength of emotion) with 5 points.

The EEG was recorded at a sampling rate of 128 Hz with 16 contact-sensors and 14 electrodes was chosen for further data analysis. The data releaser provided extracted EEG features with PSD with θ band (4-7 Hz), α band (8-13 Hz) and β band (14-30 Hz). Table 2 summarizes the characters of EEG data in DREAMER dataset.

4.2 Architecture for EEG Emotion Recognition

Assuming input data is a $N_c \times 1$ structured EEG data, where N_c is the number of channels. The EEG data first goes through a graph convolution layer with 64 Chebyshev filters. Then a 1D convolution with stride 1, kernel size 7 and 32 filters is followed. Max pooling is performed with width 2 and stride 2. The next layer is also a 1D convolution with stride 1, kernel size 7 but 16 filters. All the layers are activated by the ReLU function. After convolutions, flattening the outputs of all layers to one-dimensional vectors and concatenating together for classifying emotions. The output is a label vector containing C emotion labels. The Table 3 gives the detials about architecture of GCB-net for EEG emotion recognition.

4.3 Parameters Setting

Parameters of GCB-net are selected empirically, the number of Chebyshev polynomial parameters K is set to 4 and

TABLE 3
GCB-Net Architecture for EEG Emotion Recognition

GCB-net			
input			
$N_c \times 1$			
gcnn, 64	7×1 conv, 32	max pool, /2	7×1 conv, 16
$N_c \times 64$	$N_c \times 32$	$(N_c/2) \times 32$	$(N_c/2) \times 16$
Relu	Relu		Relu
flat			
$(N_c * 64) \times 1$		$(N_c * 16) \times 1$	$(N_c * 8) \times 1$
concatenate			
$(N_c * 88) \times 1$			
full connection		**BLS**	
connecting weights, $(N_c * 88) \times C$		feature nodes	enhancement nodes
$C \times 1$		10×10	100×10
softmax		connecting weights, $1100 \times C$	
$C \times 1$		$C \times 1$	

表 1 SEED 总结表

角色	详细信息
特征	DE PSD DASM RASM DCAU
使用电极	62 62 27 对 27 对 23 对
频率带	d; u; a; b; g 带
评分指标	正面、中性、负面

表 2 DREAMER 数据集摘要表

角色	详细信息
特性	PSD
已使用的电极	14
频率段	u; a; b 波段
评级指标	唤醒度、效价、主导性 评级量表 1, 2, 3, 4, 5

4.1 数据集

我们在 SJTU 情绪脑电数据集（SEED）[44]和 DREAMER 数据集[45]上评估了我们的方法。

4.1.1 SEED

SEED 数据采集中有 15 名参与者（7 名男性，8 名女性）。所有参与者观看了包含 5 种积极情绪、5 种中性情绪和 5 种消极情绪的 15 个电影片段。每个人在一周或更长时间间隔内参加了 3 次实验。

脑电图数据是从一个 62 通道电极帽上记录的，并降采样到 200 Hz 采样率。数据释放器通过 0.3 到 50 Hz 的带通滤波器去除了噪声和伪影。然后，应用了线性动态系统（LDS）进行进一步平滑。SEED 提供了多种提取的脑电图特征，包括差分熵（DE）、功率谱密度（PSD）、差分不对称性（DASM）、有理不对称性（RASM）和差分尾侧性（DCAU）。DE 特征和 PSD 特征提取了关于频率和能量谱的内容，DASM 特征和 RASM 特征获取了脑电图通道的不对称性信息，而 DCAU 特征计算了通道对之间的差异。每种特征都包含五个频段：δ频段（1-3 Hz）、θ频段（4-7 Hz）、α频段（8-13 Hz）、β频段（14-30 Hz）和γ频段（31-50 Hz）。SEED 的特性列于表 1。

4.1.2 DREAMER 数据集

在 DREAMER 数据集中，脑电图数据由 23 名参与者（14 名男性，9 名女性）采集。18 段电影片段被选取以唤起 9 种情绪：愉悦、兴奋、快乐、平静、愤怒、厌恶、恐惧、悲伤和惊讶。电影的时长从 65 秒到 393 秒不等，仅使用了由电影引发的 EEG 信号的最后 60 秒。观看电影后，参与者用 5 分制评估自我情绪的效价（从负到正）、唤醒度（情绪激活水平）和支配性（情绪强度）。

EEG 以 128 Hz 的采样率进行记录，选择了 16 个接触传感器和 14 个电极用于进一步的数据分析。数据提供者提供了提取的 EEG 特征，包括 u 波段（4-7 Hz）、a 波段（8-13 Hz）和 b 波段（14-30 Hz）的功率谱密度（PSD）。表 2 总结了 DREAMER 数据集中 EEG 数据的特征。

4.2 EEG 情绪识别架构

假设输入数据是一个 N 通道的 EEG 结构化数据，其中 N 是通道的数量。EEG 数据首先通过一个使用 64 个切比雪夫滤波器的图卷积层。然后进行一个步长为 1、内核大小为 7、滤波器数量为 32 的一维卷积。接着使用宽度为 2、步长为 2 的最大池化。下一层也是一个步长为 1、内核大小为 7 但滤波器数量为 16 的一维卷积。所有层都使用 ReLU 函数激活。卷积后，将所有层的输出展平为一维向量并连接起来用于情绪分类。输出是一个包含 C 个情绪标签的标签向量。表 3 给出了 GCB-net 用于 EEG 情绪识别的架构细节。

4.3 参数设置

GCB-net 的参数是经验选择，切比雪夫多项式参数的数量 K 设置为 4,

表 3 GCB-Net 架构用于脑电图情绪识别

GCB-net			
input			
$N_c \times 1$			
gcnn, 64	7×1 conv, 32	max pool, /2	7×1 conv, 16
$N_c \times 64$	$N_c \times 32$	$(N_c/2) \times 32$	$(N_c/2) \times 16$
Relu	Relu		Relu
flat			
$(N_c * 64) \times 1$		$(N_c * 16) \times 1$	$(N_c * 8) \times 1$
concatenate			
$(N_c * 88) \times 1$			
full connection		**BLS**	
connecting weights, $(N_c * 88) \times C$		feature nodes	enhancement nodes
$C \times 1$		10×10	100×10
softmax		connecting weights, $1100 \times C$	
$C \times 1$		$C \times 1$	

TABLE 4
Comparison Against Classification Accuracies/Stand Deviations (%) on SEED with SVM, DBN, GCNN, DGCNN, GCB-net and GCB-net+BLS

Feature	Model	δ band	θ band	α band	β band	γ band	all ($\delta,\theta,\alpha,\beta,\gamma$)
DE	SVM [44]	60.50/14.14	60.95/10.20	66.64/14.41	80.76/11.56	79.56/11.38	83.99/9.72
	DBN [44]	64.32/12.45	60.77/10.42	64.01/15.97	78.92/12.48	79.19/14.58	86.08/8.34
	GCNN [36]	72.75/10.85	74.40/8.23	73.46/12.17	83.24/9.93	83.36/9.43	87.40/9.20
	DGCNN [36]	74.25/11.42	71.52/5.99	74.43/12.16	83.65/10.17	85.73/10:64	90.40/8.49
	GCB-net	**80.38/10.04**	76.09/7.54	81.36/11.44	88.05/9.84	88.45/9.67	92.30/7.40
	GCB-net+BLS	79.98/8.93	**76.51/9.56**	**81.97/11.05**	**89.06/8.69**	**89.10/9.55**	**94.24/6.70**
PSD	SVM [44]	58.03/15.39	57.26/15.09	59.04/15.75	73.34/15.20	71.24/16.38	59.60/15.93
	DBN [44]	60.05/16.66	55.03/13.88	52.79/15.38	60.68/21.31	63.42/19.66	61.90/16.65
	GCNN [36]	69.89/13.83	70.92/9.18	73.18/12.74	76.21/10.76	76.15/10.09	81.31/11.26
	DGCNN [36]	71.23/11.42	71.20/8.99	73.45/12.25	77.45/10.81	76.60/11.83	81.73/9.94
	GCB-net	72.76/12.45	72.31/8.52	74.29/11.22	81.05/12.23	81.85/11.13	83.75/10.63
	GCB-net+BLS	**72.90/13.19**	**74.48/9.03**	**76.99/10.36**	**83.30/10.73**	**83.12/11.95**	**84.32/10.61**
DASM	SVM [44]	48.87/10.49	53.02/12.76	59.81/14.67	75.03/15.72	73.59/16.57	72.81/16.57
	DBN [44]	48.79/9.62	51.59/13.98	54.03/17.05	69.51/ 15.22	70.06/18.14	72.73/15.93
	GCNN [36]	57.07/6.75	54.80/9.09	62.97/13.43	74.97/13.40	73.28/13.67	76.00/13.32
	DGCNN [36]	55.93/9.14	56.12/7.86	64.27/12.72	73.61/14.35	73.50/16.6	78.45/11.84
	GCB-net	**65.04/8.18**	**65.36/8.97**	70.10/11.46	83.22/10.36	83.44/13.21	79.67/14.06
	GCB-net+BLS	62.36/10.66	65.00/10.31	**70.91/10.84**	**85.55/11.39**	**86.04/10.85**	**82.09/13.14**
RASM	SVM [44]	47.75/10.59	51.40/12.53	60.71/14.57	74.59/16.18	74.61/15.57	74.74/14.79
	DBN [44]	48.05/10.37	50.62/14.02	56.15/15.28	70.31/15.62	68.22/18.09	71.30/16.16
	GCNN [36]	59.70/5.65	55.91/8.82	59.97/14.27	79.45/13.32	79.73/13.22	84.06/12.86
	DGCNN [36]	57.79/6.90	55.79/8.10	61.58/12.63	75.79/13.07	82.32/11.54	85.00/12.47
	GCB-net	**62.66/7.22**	**65.07/8.85**	70.79/10.78	85.62/10.11	**85.79/12.05**	86.40/9.96
	GCB-net+BLS	62.56/8.83	62.22/11.12	**71.43/10.83**	**87.03/11.16**	85.59/11.18	**87.73/10.19**
DCAU	SVM [44]	55.92/14.62	57.16/10.77	61.37/15.97	75.17/15.58	76.44/15.41	77.38/11.98
	DBN [44]	54.58/12.81	56.94/12.54	57.62/13.58	70.70/16.33	72.27/16.12	77.20/14.24
	GCNN [36]	62.60/12.88	65.05/8.35	66.41/11.06	77.28/11.55	78.68/13.00	79.02/11.27
	DGCNN [36]	63.18/13.48	62.55/7.96	67.71/10.74	78.68/10.81	80.05/13.03	81.91/10.06
	GCB-net	**70.86/11.27**	69.63/9.42	70.66/10.45	82.85/11.65	82.53/11.77	83.32/9.40
	GCB-net+BLS	70.63/10.75	**69.82/9.46**	**73.43/11.38**	**87.08/9.52**	**85.56/10.64**	**86.50/8.63**

learning rate is varying for different kinds of feature input. In experiments of jointly using GCB-net and BLS, the number of groups of feature nodes n and the number of groups of enhancement nodes m are chosen to be 10 and 100 respectively. The dimension of feature nodes N_f and enhancement nodes N_e are both 10. The l_2-regularized coefficient λ of BLS is tuned in range of $10^{[-3:3]}$ in SEED and tuned in range of $10^{[-20:20]}$ in DREAMER dataset.

4.4 Experiments on SEED

4.4.1 Experimental Setting

We taken the same setting with [44] and [36]. SEED contains $15\ \text{subjects} \times 3\ \text{sessions} = 45$ experiments in total. Each experiment has 15 trials (film clips). The first 9 trials are selected to be training set and the remaining 6 trials of the same session are used as testing set. The final recognition accuracy and standard deviation are obtained form average results of $2\ \text{sessions} \times 15\ \text{subjects} = 30$ experiments.

4.4.2 Results and Analysis

In experiments, GCB-net is compared with the state-of-the-art methods, such as support vector machine (SVM) , deep belief networks (DBN) , GCNN and GDCNN. The performances of all the models are evaluated on 5 bands of DE, PSD, DASM, RASM and DCAU features. The average accuracies and standard deviations of EEG emotion recognition with these models can be viewed in Table 4. GCB-net represents the orignal network and GCB-net+BLS represents the network enhanced by BLS. The results of SVM, DBN and GCNN, DGCNN are cited from [44] and [36] respectively. Followed by [36] and [44], 2 sessions of each subject were chosen to get the final results, and the mean accuracy and standard deviation were computed from selected $15 \times 2 = 30$ experiments.

From Table 4, we can discover that the recognition accuracies of all models on β band and γ band were much higher than other bands. It may suggest that the β frequency band and γ frequency band of EEG data are more relevant to emotional responses. The performance of EEG emotion recognition on DE features exceeded other kinds of features and our models archived the best results on each band of DE features. Generally, GCB-net+BLS performed better on α band, β band, γ band and all-frequency band, and original GCB-net performed better on the rest of bands. Maybe BLS can not generate good nodes based on the feature spaces of δ band and θ band. Whatever, our models reached highest accuracies on all the bands of all the kinds of features. And GCB-net+BLS obtained the best result with the accuracy of 94.24 percent and standard deviation of 6.79 percent on all-frequency band of DE features.

表 4 在 SEED 上使用 SVM、DBN、GCNN、DGCNN、GCB-net 和 GCB-net+BLS 的分类准确率/标准差(%)比较

特征	模型	d 乐队	u 乐队	a 乐队	b 乐队	g 乐队	全部（d; u; a; b; g）
DE	SVM [44] 60.50/14.14 60.95/10.20 66.64/14.41 80.76/11.56 79.56/11.38 83.99/9.72 DBN [44] 64.32/12.45 60.77/10.42 64.01/15.97 78.92/12.48 79.19/14.58 86.08/8.34 GCNN [36] 72.75/10.85 74.40/8.23 73.46/12.17 83.24/9.93 83.36/9.43 87.40/9.20 DGCNN [36] 74.25/11.42 71.52/5.99 74.43/12.16 83.65/10.17 85.73/10:64 90.40/8.49						
	GCB-net 80:38=10:04 80:38=10:04 76.09/7.54 81.36/11.44 88.05/9.84 88.45/9.67 92.30/7.40						
	GCB-net+BLS	79.98/8.93	76:51=9:56	81:97=11:05	89:06=8:6989:06=8:69	89:10=9:55	94:24=6:7094:24=6:7
PSD	SVM [44] 58.03/15.39 57.26/15.09 59.04/15.75 73.34/15.20 71.24/16.38 59.60/15.93 DBN [44] 60.05/16.66 55.03/13.88 52.79/15.38 60.68/21.31 63.42/19.66 61.90/16.65 GCNN [36] 69.89/13.83 70.92/9.18 73.18/12.74 76.21/10.76 76.15/10.09 81.31/11.26 DGCNN [36] 71.23/11.42 71.20/8.99 73.45/12.25 77.45/10.81 76.60/11.83 81.73/9.94						
	GCB-net 72.76/12.45 72.31/8.52 74.29/11.22 81.05/12.23 81.85/11.13 83.75/10.63						
	GCB-net+BLS	72:90=13:19	74:48=9:03	76:99=10:36	83:30=10:73	83:12=11:95	84:32=10:6184:32=10:
DASM	SVM [44] 48.87/10.49 53.02/12.76 59.81/14.67 75.03/15.72 73.59/16.57 72.81/16.57 DBN [44] 48.79/9.62 51.59/13.98 54.03/17.05 69.51/ 15.22 70.06/18.14 72.73/15.93 GCNN [36] 57.07/6.75 54.80/9.09 62.97/13.43 74.97/13.40 73.28/13.67 76.00/13.32 DGCNN [36] 55.93/9.14 56.12/7.86 64.27/12.72 73.61/14.35 73.50/16.6 78.45/11.84						
	GCB-net 65:04=8:18 65:04=8:18 65:36=8:97 65:36=8:97 70.10/11.46 83.22/10.36 83.44/13.21 79.67/14.06						
	GCB-net+BLS	62.36/10.66	65.00/10.31	70:91=10:84	85:55=11:39	86:04=10:85	82:09=13:1482:09=13:
RASM	SVM [44] 47.75/10.59 51.40/12.53 60.71/14.57 74.59/16.18 74.61/15.57 74.74/14.79 DBN [44] 48.05/10.37 50.62/14.02 56.15/15.28 70.31/15.62 68.22/18.09 71.30/16.16 GCNN [36] 59.70/5.65 55.91/8.82 59.97/14.27 79.45/13.32 79.73/13.22 84.06/12.86 DGCNN [36] 57.79/6.90 55.79/8.10 61.58/12.63 75.79/13.07 82.32/11.54 85.00/12.47						
	GCB-net 62:66=7:22 62:66=7:22 65:07=8:85 65:07=8:85 70.79/10.78 85.62/10.11 85:79=12:05 85:79=12:05 86.40/9.96						
	GCB-net+BLS	62.56/8.83	62.22/11.12	71:43=10:83	87:03=11:16	85.59/11.18	87:73=10:1987:73=10:
DCAU	SVM [44] 55.92/14.62 57.16/10.77 61.37/15.97 75.17/15.58 76.44/15.41 77.38/11.98 DBN [44] 54.58/12.81 56.94/12.54 57.62/13.58 70.70/16.33 72.27/16.12 77.20/14.24 GCNN [36] 62.60/12.88 65.05/8.35 66.41/11.06 77.28/11.55 78.68/13.00 79.02/11.27 DGCNN [36] 63.18/13.48 62.55/7.96 67.71/10.74 78.68/10.81 80.05/13.03 81.91/10.06						
	GCB-net 70:86=11:27 70:86=11:27 69.63/9.42 70.66/10.45 82.85/11.65 82.53/11.77 83.32/9.40						
	GCB-net+BLS	70.63/10.75	69:82=9:46	73:43=11:38	87:08=9:5287:08=9:52	85:56=10:64	86:50=8:63

学习率因不同类型的特征输入而变化。在联合使用 GCB-net 和 BLS 的实验中，特征节点组的数量 n 和增强节点组的数量 m 分别选择为 10 和 100。特征节点 N 和增强节点 N 的维度均为 10。BLS 的 l-正则化系数在 10 的范围内进行调整。

½ § 在 SEED 和调谐范围内

$10^{½20:20}$ 在 DREAMER 数据集中。

4.4 在 SEED 上的实验

4.4.1 实验设置

我们采用了与[44]和[36]相同的设置。SEED 包含 15 个受试者 3 个试次=总共 45 个实验。每个实验有 15 次试验（电影片段）。前 9 次试验被选为训练集，同一试次的剩余 6 次试验用作测试集。最终识别准确率和标准差是通过 2 个试次 15 个受试者=30 个实验的平均结果获得的。

4.4.2 结果与分析

在实验中，GCB-net 与最先进的方法进行了比较，例如支持向量机（SVM）、深度信念网络（DBN）、GCNN 和 GDCNN。所有模型的性能都在 DE、PSD、DASM 的 5 个波段上进行了评估。

RASM 和 DCAU 功能。使用这些模型进行脑电图情绪识别的平均准确率和标准差可以查看在表 4 中。GCB-net 代表原始网络，GCB-net+BLS 代表通过 BLS 增强的网络。SVM、DBN 和 GCNN、DGCNN 的结果分别引自[44]和[36]。根据[36]和[44]，每个被试选择了 2 个会话来获得最终结果，从选定的 15×2=30 个实验中计算了平均准确率和标准差。

从表 4 中，我们可以发现所有模型在 b 波段和 g 波段上的识别准确率都远高于其他波段。这可能表明 EEG 数据的 b 频段和 g 频段与情绪反应更相关。在 DE 特征上的脑电波情绪识别性能超过了其他类型的特征，并且我们的模型在 DE 特征的每个波段上都取得了最佳结果。通常，GCB-net+BLS 在 a 波段、b 波段、g 波段和全频段上表现更好，而原始 GCB-net 在其他波段上表现更好。也许 BLS 无法基于 d 波段和 u 波段的特征空间生成良好的节点。无论如何，我们的模型在所有类型特征的各个波段上达到了最高的准确率。并且 GCB-net+BLS 在全频段的 DE 特征上获得了最佳结果，准确率为 94.24%，标准差为 6.79%。

TABLE 5
Comparision Against Classification Accuracies/Stand Deviations (%) on DREAMER Dataset with SVM, GraphSLDA, GSCCA, DGCNN, GCB-net and GCB-net+BLS

Model	Valence	Arousal	Dominance
SVM [36]	60.14/33.34	68.84/24.94	75.84/20.76
GraphSLDA [36]	57.70/13.89	68.12/17.53	73.90/15.85
GSCCA [36]	56.65/21.50	70.30/18.66	77.31/15.44
DGCNN [36]	86.23/12.29	84.54/10.18	85.02/10.25
GCB-net (SR)	**86.99/6.21**	**89.32/5.01**	**89.20/4.33**
GCB-net (BLS)	85.98/7.91	87.48/6.48	88.21/5.77

4.5 Experiments on DREAMER Dataset

4.5.1 Experimental Setting

For comparing, the same experimental setting is also taken as [36]. The subject-dependent leave-one-session-out cross-validation strategy was chosen to evaluate performance of models. For each subject, 18 film clips were watched and corresponding EEG data of 18 sessions were recorded. In one experiment, data in 17 sessions of the same subject were used for training and the data in rest session were used for testing. Repeating 18 times, when data in each sessions were treated as the testing data, the result of each subject can be obtained by averaging the recognition accuracies of all session times. And the final recognition accuracy of experiments can be computed by averaging the results of 23 subjects. To implement binary classifications, 5 points are divided into 2 classes: first 2 rating scales belong to the low arousal, valence or dominance and last 3 rating scales belong ro the high arousal, valence or dominance.

4.5.2 Results and Analysis

To evaluate the performance of GCB-net on DREAMER dataset, our methods were compared with SVM, graph regularized sparse linear discriminant analysis (GraphSLDA) [46], group sparse canonical correlation analysis (GSCCA) [47] and DGCNN. All the experiments were conducted on PSD features of all-frequency band (θ, α, β). And Table 5 gives average recognition accuracies and standard deviations of all the models on dimensions of Valence, Arousal, Dominance. The results of SVM, GraphSLDA and GSCCA are cited from [36].

From Table 5 we can see that most methods performed better on Dominance dimension than Valence and Arousal dimensions. It may suggest Dominance dimension is easier to predict, though most researcher preferred Valence and Arousal dimension. From the aspect of model, GCB-net achieved higher recognition accuracies than other models on all dimensions with same setting, which demonstrated the superiority of feature construction for GCB-net. We can observe that GCB-net+BLS obtained highest accuracies of 86.99, 89.32 and 89.20 percent on dimensions of Valence, Arousal and Dominance respectively. Although, the performance of GCB-net+BLS has no promotion with GCB-net, the accuracies are still better or close to the results of DGCNN, implying the effectiveness of GCB feature.

5 Conclusion

In this paper, GCB-net was introduced to recognize emotions with EEG-channel signals. It is a network which can explore the deeper-level information of graph-structured data. This paper first uses the graph convolutional layer to deal with the graph-structured input and then stacks multiple regular CNN layers to abstract high-level features. Finally, outputs of all hierarchical layers are connected to provide the model with broad searching spaces. In the experiments of SEED, the GCB-net was compared with SVM, DBN, GCNN and DGCNN. The proposed GCB-net +BLS achieved the highest accuracy of 94.24 percent on the DE feature of the all-frequency band. Occasionally the performance the GCB-net with BLS was poorer than that of the original GCB-net, but the accuracy of GCB-net + BLS was still higher than that of other methods, thus proving that the feature learned using the GCB-net is feasible and distinguished. In the experiments of DREAMER dataset, the GCB-net was compared with SVM, GraphSLDA, GSCCA and DGCNN. The proposed GCB-net obtained accuracies of 86.99, 89.32 and 89.20 percent on dimensions of Valence, Arousal and Dominance respectively, which were higher than other models with the same setting. The experimental results demonstrated the robust classifying ability of the GCB-net and BLS in EEG emotion recognition.

Acknowledgments

The authors would like to thank the support of Key Laboratory of Child Development and Learning Science (Southeast University), Brain-like Computing and Machine Intelligence (BCMI) Laboratory and University of the West of Scotland, School of Engineering and Computing. This work was supported in part by the National Natural Science Foundation of China grant under number 61702195, 61751202, U1813203, 61572540, U1801262.

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表 5 在 DREAMER 数据集上使用 SVM、GraphSLDA 的分类准确率/标准偏差比较 (%)

GSCCA、DGCNN、GCB-net 和 GCB-net+BLS

模型	价态	唤醒	支配
SVM [36]	60.14/33.34	68.84/24.94	75.84/20.76
GraphSLDA [36]	57.70/13.89	68.12/17.53	73.90/15.85
GSCCA [36]	56.65/21.50	70.30/18.66	77.31/15.44
DGCNN [36]	86.23/12.29	84.54/10.18	85.02/10.25
GCB-net (SR)	86:99=6:21 86:99=6:21	89:32=5:01 89:32=5:01	89:20=4:33
GCB-net (BLS)	85.98/7.91	87.48/6.48	88.21/5.77

4.5 DREAMER 数据集上的实验

4.5.1 实验设置

为了比较，也采用了与[36]相同的实验设置。选择受试者相关的留一会话交叉验证策略来评估模型的性能。对于每个受试者，观看 18 个电影片段并记录相应的 18 会话的脑电图数据。在一个实验中，使用同一受试者的 17 会话数据用于训练，而剩余会话的数据用于测试。重复 18 次，当每个会话的数据被用作测试数据时，通过平均所有会话时间的识别准确率可以得到每个受试者的结果。最终实验的识别准确率可以通过平均 23 个受试者的结果来计算。为了实现二元分类，5 个点被分为 2 个类别：前 2 个评分量表属于低唤醒、效价或支配，最后 3 个评分量表属于高唤醒、效价或支配。

4.5.2 结果与分析

为了评估 GCB-net 在 DREAMER 数据集上的性能，我们的方法与 SVM、图正则化稀疏线性判别分析（GraphSLDA）[46]、组稀疏典型相关分析（GSCCA）[47]和 DGCNN 进行了比较。所有实验均在所有频段（u；a；b）的 PSD 特征上进行。表 5 给出了所有模型在效价、唤醒度和支配性维度上的平均识别准确率和标准差。SVM、GraphSLDA 和 GSCCA 的结果引自[36]。

从表 5 我们可以看到，大多数方法在支配维度上的表现优于效价和唤醒维度。这可能表明支配维度更容易预测，尽管大多数研究者更倾向于效价和唤醒维度。从模型的角度来看，在相同设置下，GCB-net 在所有维度上实现了比其他模型更高的识别准确率，这证明了 GCB-net 特征构建的优越性。我们可以观察到，GCB-net+BLS 在效价、唤醒和支配维度上分别获得了最高的准确率，分别为 86.99%、89.32%和 89.20%。尽管 GCB-net+BLS 的性能与 GCB-net 没有提升，但准确率仍然优于或接近 DGCNN 的结果，这表明了 GCB 特征的有效性。

5 结论

在本文中，介绍了 GCB-net 用于识别脑电图通道信号的情绪这是一个可以探索图结构数据深层信息的网络。本文首先使用图卷积层处理图结构输入，然后堆叠多个常规 CNN 层来提取高级特征。最后，将所有层级层的输出连接起来，为模型提供广阔的搜索空间。在 SEED 实验中，GCB-net 与 SVM、DBN、GCNN 和 DGCNN 进行了比较。提出的 GCB-net+BLS 在全频段 DE 特征上达到了 94.24%的最高准确率。偶尔 BLS 的 GCB-net 性能比原始 GCB-net 差，但 GCB-net+BLS 的准确率仍高于其他方法，从而证明了使用 GCB-net 学习到的特征是可行且出色的。在 DREAMER 数据集实验中，GCB-net 与 SVM、GraphSLDA、GSCCA 和 DGCNN 进行了比较。提出的 GCB-net 在效价、唤醒度和主导性维度上分别获得了 86.99%、89.32%和 89.20%的准确率，这些结果高于其他设置相同的模型。实验结果证明了 GCB-net 和 BLS 在脑电图情绪识别中的强大分类能力。

致谢

作者感谢儿童发展与学习科学重点实验室（东南大学）、类脑计算与机器智能实验室（BCMI）以及西苏格兰大学工程学院和计算学院的资助。这项工作部分得到了中国国家自然科学基金资助，项目编号为 61702195、61751202、U1813203、61572540、U1801262。

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